

Selective Repeat (SR) Protocol

(A High-Efficiency Sliding Window Approach to Reliable Data Transfer)

1 Introduction

The **Selective Repeat (SR)** protocol is an advanced version of the **Go-Back-N (GBN)** protocol that improves efficiency in high-delay or high-error networks. While GBN retransmits all packets after a loss, SR retransmits **only the specific packets that are lost or corrupted**.

Key Concept

Each packet is handled independently—acknowledged, buffered, and retransmitted separately. This prevents unnecessary retransmissions and increases channel utilization.

In networks with large bandwidth-delay products, SR performs much better than GBN by maintaining multiple outstanding packets while avoiding redundant retransmissions.

2 Sender and Receiver Window Concept

2.1 Sender Window

The sender can transmit up to N unacknowledged packets at a time. Each packet maintains its own timer and acknowledgment state.

- **send_base**: sequence number of the oldest unacknowledged packet.
- **nextseqnum**: sequence number to be assigned to the next outgoing packet.

2.2 Receiver Window

The receiver maintains a window of size N and can accept out-of-order packets. Packets received correctly are buffered until all preceding packets are delivered.

- **rcv_base**: sequence number of the next expected in-order packet.
- **Buffered Packets**: out-of-order packets that have been received correctly.

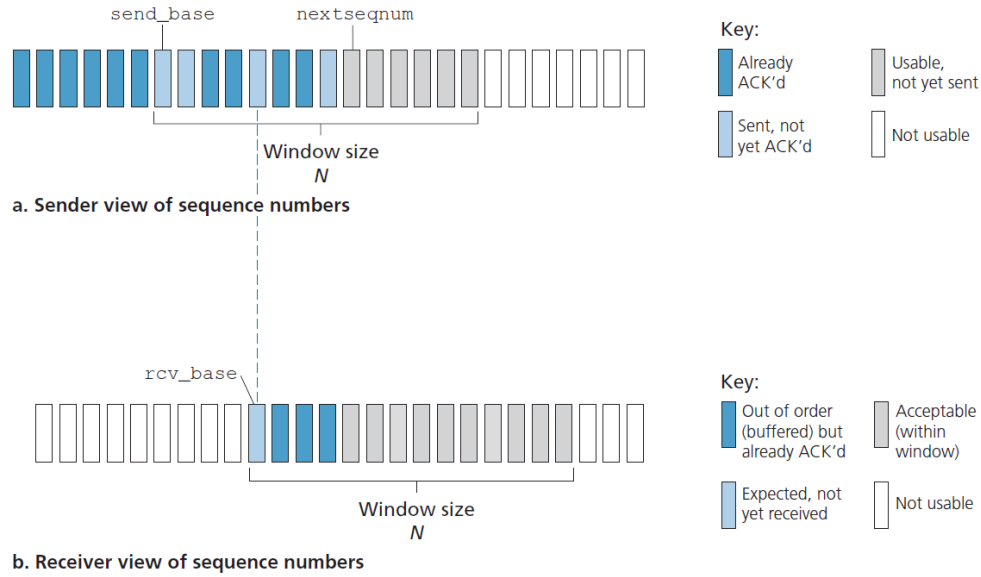


Figure 1: Sender's and Receiver's Sliding Window and Buffering in SR

GBN vs SR

- **GBN:** Discards out-of-order packets.
- **SR:** Buffers out-of-order packets until missing ones arrive.

3 Protocol Operation

3.1 Sender Behavior

The SR sender performs three main actions:

1. **Data from Upper Layer:** If the sequence number is within the sender's window:

$$\text{sndpkt}[i] = \text{make_pkt}(i, \text{data}, \text{checksum})$$

The packet is transmitted and a timer is started. Otherwise, the data is buffered for later transmission.

2. **Timeout Event:** If a packet's timer expires, that specific packet is retransmitted. Each packet thus has its own logical timer.
3. **ACK Received:** When an acknowledgment for packet i arrives:
 - Mark packet i as acknowledged.
 - If $i = \text{send_base}$, move the base forward to the next unacknowledged packet.
 - Transmit new packets if the window has space.

3.2 Receiver Behavior

The receiver's logic:

1. **In-Window Packet Received:** If sequence $\in [rcv_base, rcv_base + N - 1]$ and not corrupted:
 - Send an ACK immediately.
 - If not previously received, buffer it.
 - If it matches rcv_base , deliver it and consecutive buffered packets.
 - Move receive window forward.
2. **Old Packet Received:** If sequence $\in [rcv_base - N, rcv_base - 1]$, re-ACK to ensure sender window progression.
3. **Otherwise:** Ignore the packet.

Why Re-ACK?

Reacknowledgment of previously acknowledged packets is essential — it ensures sender window movement even if certain ACKs are lost.

4 Example: SR Operation

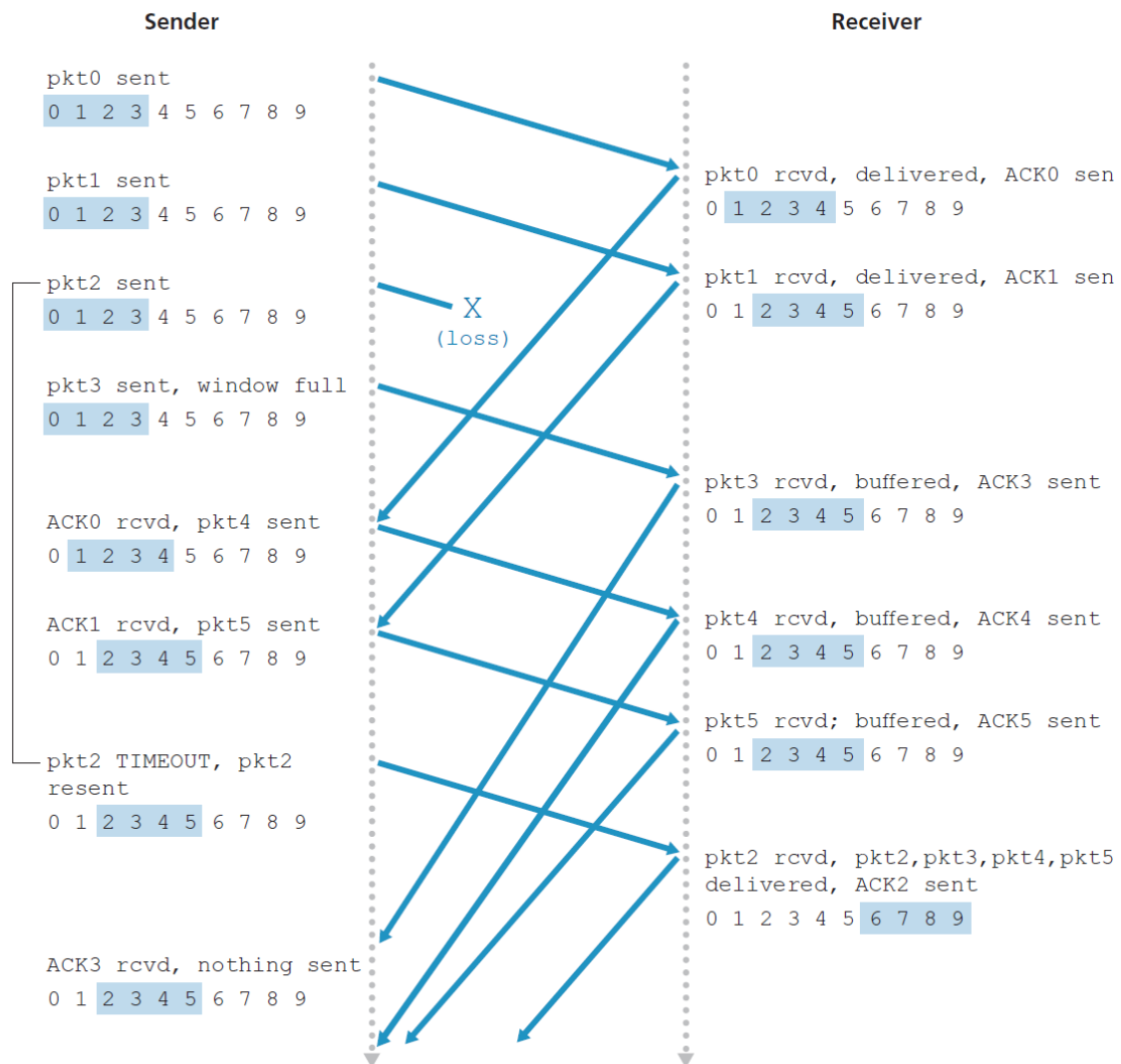


Figure 2: Selective Repeat Operation Example

- Packets 0–1 are received and delivered normally.
- Packets 3–5 are received and buffered (packet 2 missing).
- When packet 2 arrives, packets 2–5 are delivered together.
- Only packet 2 was retransmitted — avoiding unnecessary retransmissions.

5 Sequence Number Dilemma

In SR, sender and receiver windows may not always align, leading to ambiguity when sequence numbers wrap around.

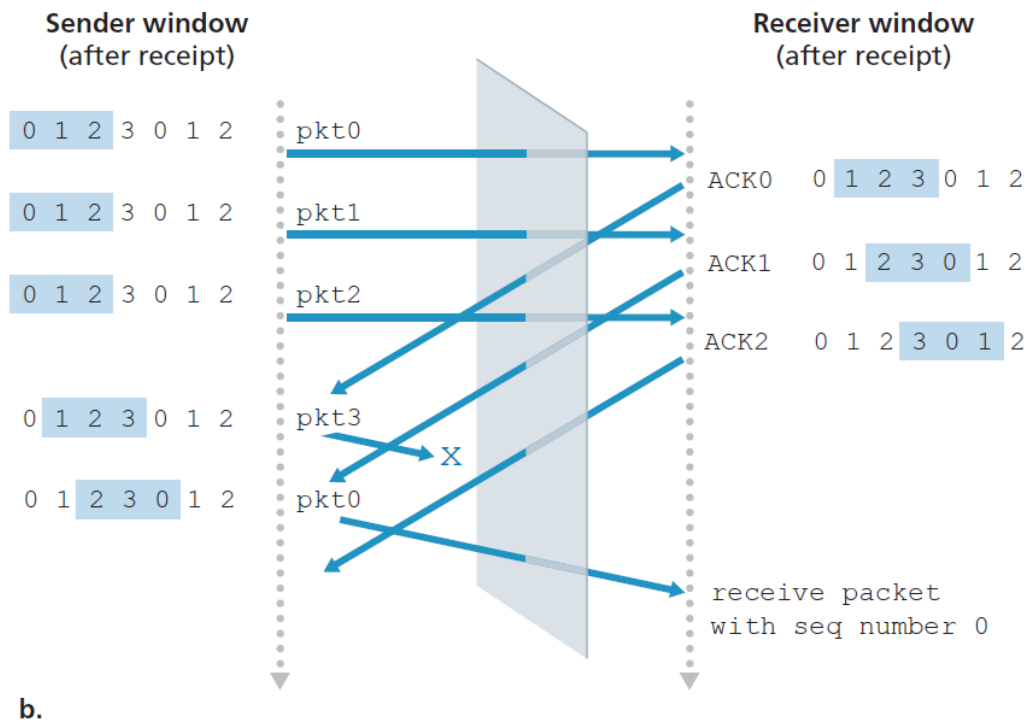
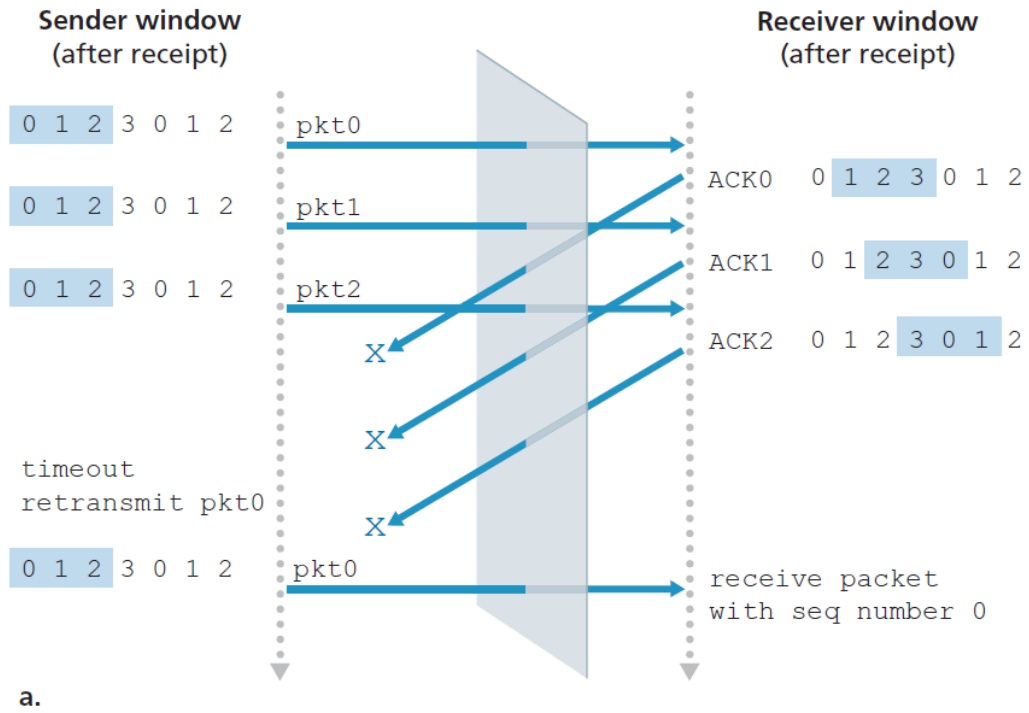


Figure 3: SR Receiver Dilemma with Finite Sequence Numbers

Important Condition

For correct SR operation:

$$N \leq \frac{K}{2}$$

where K is the total number of unique sequence numbers.

If $N > K/2$, the receiver may confuse old retransmissions with new packets — leading to data duplication or corruption.

6 Summary

Selective Repeat Highlights

- Retransmits only specific lost/corrupted packets.
- Receiver buffers out-of-order packets.
- Each packet has its own timer and ACK.
- Higher efficiency than Go-Back-N, especially in noisy or high-delay links.

Advantages:

- Eliminates unnecessary retransmissions.
- Maximizes link utilization for high-delay networks.

Disadvantages:

- More complex (requires per-packet timers and buffers).
- Requires a larger sequence number space.

In summary: The Selective Repeat protocol strikes a balance between reliability and efficiency. It forms the foundation for modern transport mechanisms like **TCP with Selective Acknowledgment (SACK)**.